

Shawn Guo

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EDUCATION

Carnegie Mellon University

M.S. in Software Engineering

MOUNTAIN VIEW, CA

Expected December 2027

Johns Hopkins University

B.S. in Computer Science; B.M. in Music for New Media | GPA: 3.91/4.00

BALTIMORE, MD

May 2026

- Relevant Courses: Computer Graphics and 3D Game Programming, Linear Algebra, Algorithms, Artificial Intelligence

EXPERIENCE

Software Engineering Intern, Audio

Activision/Raven Software

MIDDLETON, WI

May 2025 - August 2025

- Developed low-level C/C++ runtime systems and audio APIs in a proprietary AAA engine for Call of Duty
- Extended a node-based, sample-aware character-dialogue system, adding negative post-node delays, client-mask targeting, and visual debug tooling. Optimized dialogue systems to cut conversations memory use by approx. 40%
- Built C#/.NET WPF audio tooling for compression, legacy CSV migration, and VO-driven facial animation

Software Engineer, Audio

Snow Day Games LLC

BALTIMORE, MD

March 2025 - Present

- Co-founded Snow Day Games and lead audio development for Vesuvius, a \$15K grant-funded action-adventure
- Integrated Wwise with gameplay systems for ambience, combat, UI, and cutscenes in Unity
- Built a data-driven audio layer across 50+ event/state controls, reducing runtime audio lookups by approx. 30%

Systems and 3D Audio Programmer

Serious Point Games Ltd

REMOTE

March 2024 - May 2025

- Designed and implemented a data-driven event architecture powering 90%+ of gameplay, UI, and audio interactions
- Integrated Wwise Rooms & Portals spatial audio for room-aware reverb and dynamic propagation
- Reduced spatial-audio memory overhead by approx. 30% and lowered room reverb authoring time by approx. 40%

Full Stack Engineer

Quest2Learn LLC

BALTIMORE, MD

December 2023 - October 2024

- Built React Three Fiber learning modules and migration pipelines for a mobile/web AR education platform
- Developed reusable 3D UI, scene logic, and web interaction systems
- Migrated 40% of legacy Unity content into web-based workflows using JSON schemas

PROJECTS

Negative Space – Spatial Audio in a Custom C++ Engine | C++, raylib, DSP

December 2025 - Present

- Built a custom C++ spatial audio stack for the Negative Space engine to a full 10+ feature runtime audio layer, including 3D emitters/listeners, partitioned convolution reverb, attenuation, panning, and limiting
- Implemented room-aware propagation, obstruction/diffraction traces, and JSON-authored acoustic zones
- Optimized audio work out of the frame update path to reduce frame drops by 75%+ compared to first iteration

Lossy Audio Lab | C++, Dear ImGui, miniaudio, Opus, DSP

March 2026 - June 2026

- Built a desktop tool to simulate and audition packet loss, jitter, Opus FEC/PLC, and redundancy on WAV/MP3 audio
- Implemented 10 ms frame packetization, jitter buffering, and recovery timeline visualization
- Profiled SIMD DSP paths, accelerating peak detection by 24.5x with NEON on Apple Silicon

TEACHING AND LEADERSHIP

Head TA | JHU Game Engines Programming (601.355) and Computer Graphics & 3D Game Programming (601.257)

President and Co-founder | Hopkins Game Development Society

SKILLS

Languages | C/C++, C#, Python, Java, Lua, TypeScript/JavaScript, HTML/CSS

Frameworks | React, Node.js, Tailwind, React Three Fiber, Unity, Unreal Engine 5

Tools & Middleware | Wwise, FMOD, Dear ImGui, .NET/WPF, Reaper, Pro Tools, Git, Perforce